Edge-Deployable AI Platform for AI-Based NeuroFusion for EEG Abnormality Measurement

Department of Electronics & Communication Engineering

Project Phase I | Guide: Ms. Harsha A

Team : **Shine Daniel · Roshan V N · Seban Jais · Naveen Jose Kurian**

Introduction

50 Million

people worldwide live with epilepsy, making it one of the most common neurological disorders (WHO)

30 Minutes

preictal window before a seizure — the critical early-warning opportunity that current systems miss

The Challenge

Current EEG diagnosis relies on manual expert interpretation — slow, subjective, and unavailable in most clinical settings. Cloud-based AI solutions introduce latency and data privacy risks that are unacceptable for continuous medical monitoring.

Our Approach

We build an **edge-deployable AI platform** that acquires EEG data via a clinical-grade ADC, conditions the signal on-device, and runs a Temporal Convolutional Network (TCN) locally — zero cloud dependency, real-time output, full patient privacy.

Objectives

01 Interface the TI ADS1299EEGFE-PDK evaluation board for clinical-grade 24-bit EEG data acquisition across N channels via SPI.

02 Implement signal conditioning to produce a clean NxM data matrix.

03 Train and validate a TCN model for 4-state EEG classification: Interictal, Preictal, Ictal, and Postictal.

04 Optimize the model for edge deployment on a Raspberry Pi 5.

05 Do better seizure prediction.

Problem Statement

🏥 Clinical Bottleneck

Manual EEG interpretation requires specialist neurologists and is unavailable in real time. Seizures can begin 30 minutes after preictal onset — early detection is critical but currently impractical at scale.

☁️ Cloud Dependency

Existing AI EEG solutions rely on cloud servers, introducing 200–500 ms latency, data privacy risks, and failure during connectivity loss — unacceptable for medical monitoring.

⚡ Compute Constraints

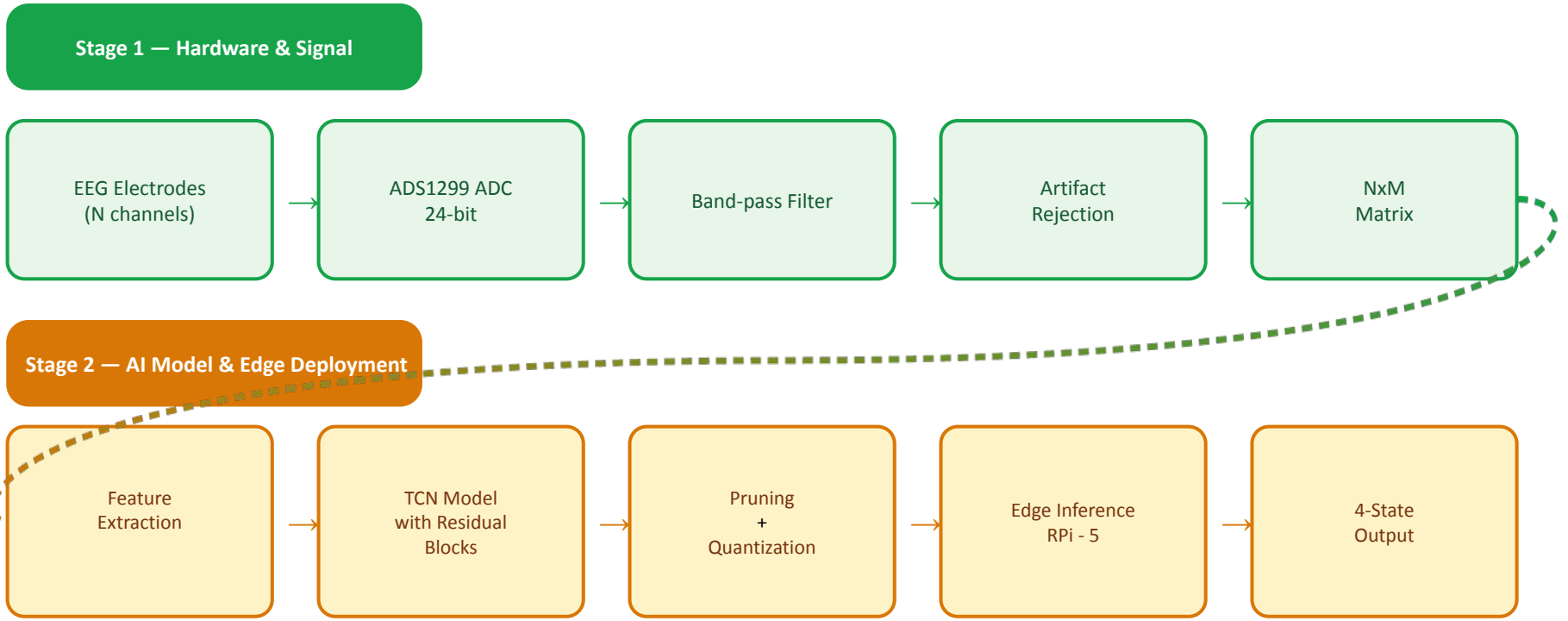
Deep learning models (TCN, LSTM) are computationally expensive. Deploying them on low-power edge devices requires model optimization without sacrificing diagnostic accuracy.

📡 Signal Noise & Quality

Raw EEG signals are contaminated by electrode artifacts, muscle movement, and power line interference. Robust preprocessing is required before AI inference.

Proposed Solution

A two-stage pipeline bridging hardware signal acquisition with edge AI inference:



Hardware & Software Requirements

🔧 Hardware

Component	Specification
EEG ADC	ADS1299EEGFE-PDK (TI)
Electrodes	N-channel scalp EEG
Edge Platform	RPi 5
Interface	SPI
Power	5V, 2–4A

💻 Software

Component	Tools / Frameworks
AI Framework	PyTorch / TensorFlow
Edge Runtime	TensorRT / TFLite
Signal Processing	SciPy, MNE-Python
Dataset	CHB-MIT / Bonn EEG
Languages	Python, C++ (inference)

Work Division

Team 1 — Hardware & Signal

- ADS1299 board interfacing & SPI communication setup
- Analog front-end design & electrode preparation
- Band-pass filter implementation
- Artifact rejection & lead-off detection handling
- Produce validated NxM data matrix from real EEG hardware

Team 2 — AI & Deployment

- TCN model architecture design & training on CHB-MIT/Bonn dataset
- Software simulation using the NxM data matrix format
- Model optimization — pruning + quantization
- Edge deployment on a Raspberry Pi 5
- Real-time inference demo, benchmarking & accuracy validation

Integration Point: Team 1's NxM hardware matrix plugs directly into Team 2's AI pipeline — same format as the software simulation.

Conclusion

Key Contributions

- Complete pipeline from raw EEG electrodes to AI classification on edge hardware
- Clinical-grade ADS1299 interfacing producing a validated N - channel data stream
- TCN model with better accuracy
- Full model optimization pipeline: pruning + quantization for edge deployment

Future Scope

- Expand to more EEG channels for higher spatial resolution
- Real-time clinical validation with patient-specific calibration
- Wearable form factor integration for continuous home monitoring
- Federated learning for multi-patient model improvement

Thank You

Seminar Topics

[1] S. Al-Marzouki, "Advancing epileptic seizure recognition through bidirectional LSTM networks," Frontiers in Computational Neuroscience, vol. 19, 2025, Art. no. 1668358. doi: 10.3389/fncom.2025.1668358.

[2] P. A. Akor, G. Enemali, U. Muhammad, R. R. Singh, and H. Larijani, "An attention-residual convolutional network for real-time seizure classification on edge devices," Sensors, vol. 25, no. 22, Art. no. 6855, Nov. 2025. doi: 10.3390/s25226855.

[3] Z. Y. Lim, Y. H. Pang, S. Y. Ooi, W. H. Khoh, and Y. J. Chew, "NeuroFed-LightTCN: Federated lightweight temporal convolutional networks for privacy-preserving seizure detection in EEG data," Applied Sciences, vol. 15, no. 17, Art. no. 9660, Sep. 2025. doi: 10.3390/app15179660.

[4] L. Huang, K. Zhou, S. Chen, Y. Chen, and J. Zhang, "Automatic detection of epilepsy from EEGs using a temporal convolutional network with a self-attention layer," BioMedical Engineering OnLine, vol. 23, Art. no. 50, Jun. 2024. doi: 10.1186/s12938-024-01244-w

References

[1] P. Zhu, W. Zhou, C. Cao, G. Liu, Z. Liu, and W. Shang, "A Novel SE-TCN-BIGRU Hybrid Network for Automatic Seizure Detection," *IEEE Access*, vol. 11, 2023, doi: 10.1109/ACCESS.2024.3406909.

[2] R. M. a. A. Leema, “Efficient Epileptic Seizure Detection with Optimal Channel Selection and FIXUPPACTBI-LSTM Deep Learning Model,” *IJTech - International Journal of Technology*. https://ijtech.eng.ui.ac.id/article/view/7333?hl=en-GB

[3] "Machine Learning Applied to Edge Computing and Wearable Devices for Healthcare: Systematic Mapping of the Literature," Sensors, vol. 24, no. 19, 2024. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC11478846/

[4] "Global, Regional, and National Burden of Epilepsy, 1990–2021: A Systematic Analysis for the Global Burden of Disease Study 2021," The Lancet Public Health, 2025. doi: 10.1016/S2468-2667(24)00302-5. [Online]. Available: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11876103/

[5] J. P. K. Jayalakshmi and P. Manimegalai, "Residual TCN-BiLSTM with attention framework for epileptic seizure detection from EEG signals," Int. J. Intell. Eng. Syst., vol. 19, no. 7, pp. 28–48, 2026. doi: 10.22266/ijies2026.0731.03.